In [90]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

In [155…
```
df = pd.read_csv(r"C:\Project DS/loan.csv")
```

```
C:\Users\pulsa\AppData\Local\Temp\ipykernel_14312\2152967141.py:1: DtypeWarning:
Columns (19,55) have mixed types. Specify dtype option on import or set low_memor
y=False.
  df = pd.read_csv(r"C:\Project DS/loan.csv")
```

In [156…
```
dic = pd.read_excel(r"C:\Project DS/LCDataDictionary.xlsx")
```

```
C:\Users\pulsa\anaconda3\Lib\site-packages\openpyxl\worksheet\_reader.py:329: Use
rWarning: Unknown extension is not supported and will be removed
  warn(msg)
```

In [157…
```
df = pd.read_csv(
    r"C:\Project DS\loan.csv",
    dtype={19: str, 55: str}
)
```

In [158…
```
for col in df.columns:
    if df[col].apply(type).nunique() > 1:
        print(col)
```

```
emp_title
emp_length
desc
title
earliest_cr_line
last_pymnt_d
next_pymnt_d
last_credit_pull_d
verification_status_joint
```

In [159…
```
for i, col in enumerate(df.columns):
    print(i, col)
```

```
0 id
1 member_id
2 loan_amnt
3 funded_amnt
4 funded_amnt_inv
5 term
6 int_rate
7 installment
8 grade
9 sub_grade
10 emp_title
11 emp_length
12 home_ownership
13 annual_inc
14 verification_status
15 issue_d
16 loan_status
17 pymnt_plan
18 url
19 desc
20 purpose
21 title
22 zip_code
23 addr_state
24 dti
25 delinq_2yrs
26 earliest_cr_line
27 inq_last_6mths
28 mths_since_last_delinq
29 mths_since_last_record
30 open_acc
31 pub_rec
32 revol_bal
33 revol_util
34 total_acc
35 initial_list_status
36 out_prncp
37 out_prncp_inv
38 total_pymnt
39 total_pymnt_inv
40 total_rec_prncp
41 total_rec_int
42 total_rec_late_fee
43 recoveries
44 collection_recovery_fee
45 last_pymnt_d
46 last_pymnt_amnt
47 next_pymnt_d
48 last_credit_pull_d
49 collections_12_mths_ex_med
50 mths_since_last_major_derog
51 policy_code
52 application_type
53 annual_inc_joint
54 dti_joint
55 verification_status_joint
56 acc_now_delinq
57 tot_coll_amt
58 tot_cur_bal
59 open_acc_6m
```

```
60 open_il_6m
61 open_il_12m
62 open_il_24m
63 mths_since_rcnt_il
64 total_bal_il
65 il_util
66 open_rv_12m
67 open_rv_24m
68 max_bal_bc
69 all_util
70 total_rev_hi_lim
71 inq_fi
72 total_cu_tl
73 inq_last_12m
```

In [160…

```python
df = pd.read_csv(
    r"C:\Project DS\loan.csv",
    dtype={
        "desc": str,
        "verification_status_joint": str
    },
    low_memory=False
)
```

In [161…

```python
df.shape
```

Out[161…

```
(887379, 74)
```

In [162…

```python
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 887379 entries, 0 to 887378
Data columns (total 74 columns):
 #   Column                       Non-Null Count   Dtype  
---  ------                       --------------   -----  
 0   id                           887379 non-null  int64  
 1   member_id                    887379 non-null  int64  
 2   loan_amnt                    887379 non-null  float64
 3   funded_amnt                  887379 non-null  float64
 4   funded_amnt_inv              887379 non-null  float64
 5   term                         887379 non-null  object 
 6   int_rate                     887379 non-null  float64
 7   installment                  887379 non-null  float64
 8   grade                        887379 non-null  object 
 9   sub_grade                    887379 non-null  object 
 10  emp_title                    835917 non-null  object 
 11  emp_length                   842554 non-null  object 
 12  home_ownership               887379 non-null  object 
 13  annual_inc                   887375 non-null  float64
 14  verification_status          887379 non-null  object 
 15  issue_d                      887379 non-null  object 
 16  loan_status                  887379 non-null  object 
 17  pymnt_plan                   887379 non-null  object 
 18  url                          887379 non-null  object 
 19  desc                         126026 non-null  object 
 20  purpose                      887379 non-null  object 
 21  title                        887226 non-null  object 
 22  zip_code                     887379 non-null  object 
 23  addr_state                   887379 non-null  object 
 24  dti                          887379 non-null  float64
 25  delinq_2yrs                  887350 non-null  float64
 26  earliest_cr_line             887350 non-null  object 
 27  inq_last_6mths               887350 non-null  float64
 28  mths_since_last_delinq       433067 non-null  float64
 29  mths_since_last_record       137053 non-null  float64
 30  open_acc                     887350 non-null  float64
 31  pub_rec                      887350 non-null  float64
 32  revol_bal                    887379 non-null  float64
 33  revol_util                   886877 non-null  float64
 34  total_acc                    887350 non-null  float64
 35  initial_list_status          887379 non-null  object 
 36  out_prncp                    887379 non-null  float64
 37  out_prncp_inv                887379 non-null  float64
 38  total_pymnt                  887379 non-null  float64
 39  total_pymnt_inv              887379 non-null  float64
 40  total_rec_prncp              887379 non-null  float64
 41  total_rec_int                887379 non-null  float64
 42  total_rec_late_fee           887379 non-null  float64
 43  recoveries                   887379 non-null  float64
 44  collection_recovery_fee      887379 non-null  float64
 45  last_pymnt_d                 869720 non-null  object 
 46  last_pymnt_amnt              887379 non-null  float64
 47  next_pymnt_d                 634408 non-null  object 
 48  last_credit_pull_d           887326 non-null  object 
 49  collections_12_mths_ex_med   887234 non-null  float64
 50  mths_since_last_major_derog  221703 non-null  float64
 51  policy_code                  887379 non-null  float64
 52  application_type             887379 non-null  object 
 53  annual_inc_joint             511 non-null     float64
 54  dti_joint                    509 non-null     float64
```

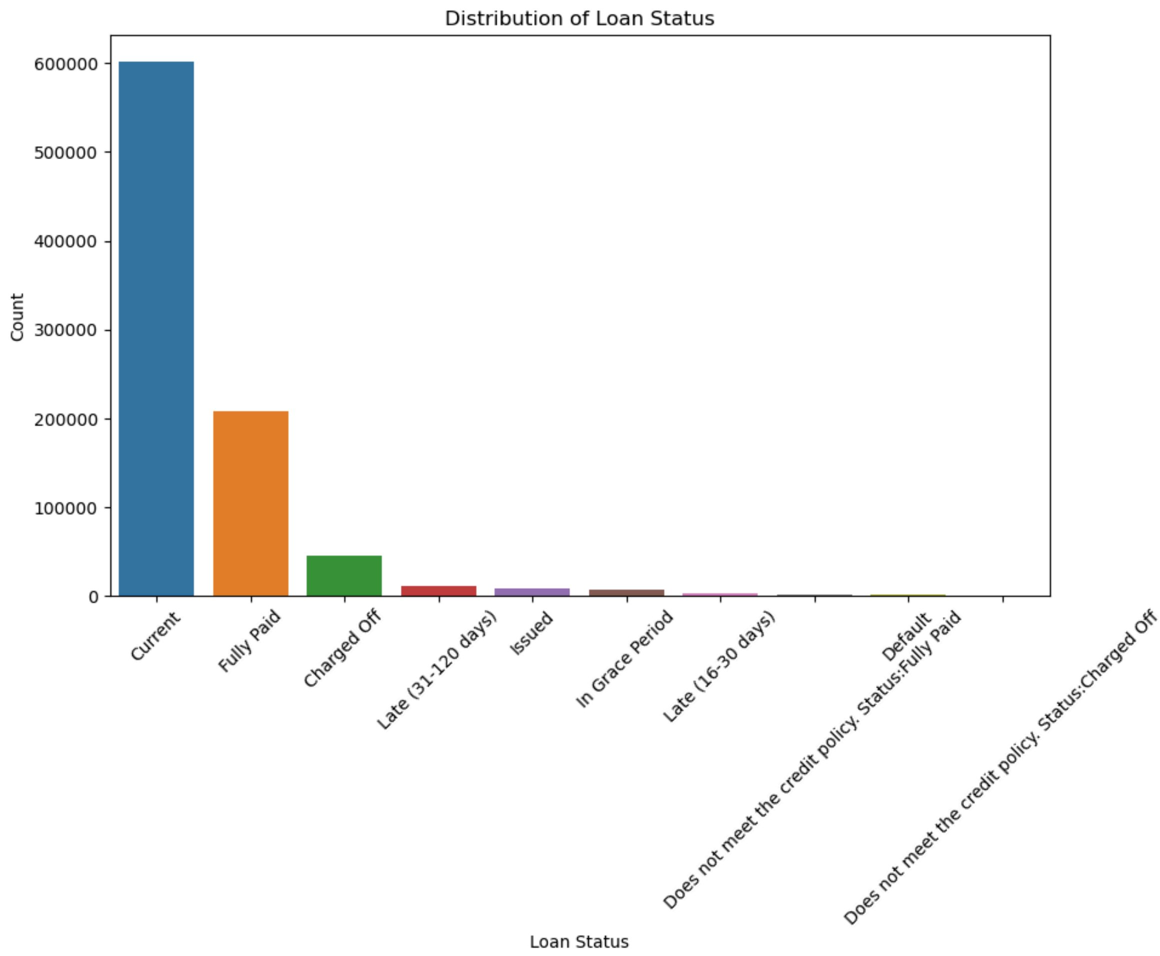
```
 55  verification_status_joint    511 non-null     object
 56  acc_now_delinq               887350 non-null  float64
 57  tot_coll_amt                 817103 non-null  float64
 58  tot_cur_bal                  817103 non-null  float64
 59  open_acc_6m                  21372 non-null   float64
 60  open_il_6m                   21372 non-null   float64
 61  open_il_12m                  21372 non-null   float64
 62  open_il_24m                  21372 non-null   float64
 63  mths_since_rcnt_il           20810 non-null   float64
 64  total_bal_il                 21372 non-null   float64
 65  il_util                      18617 non-null   float64
 66  open_rv_12m                  21372 non-null   float64
 67  open_rv_24m                  21372 non-null   float64
 68  max_bal_bc                   21372 non-null   float64
 69  all_util                     21372 non-null   float64
 70  total_rev_hi_lim             817103 non-null  float64
 71  inq_fi                       21372 non-null   float64
 72  total_cu_tl                  21372 non-null   float64
 73  inq_last_12m                 21372 non-null   float64
dtypes: float64(49), int64(2), object(23)
memory usage: 501.0+ MB
```

In [163…

```
import matplotlib.pyplot as plt
import seaborn as sns

status_counts = df['loan_status'].value_counts()

# 绘制柱状图
plt.figure(figsize=(10, 6))
sns.barplot(x=status_counts.index, y=status_counts.values)
plt.title('Distribution of Loan Status')
plt.xlabel('Loan Status')
plt.ylabel('Count')
plt.xticks(rotation=45)
plt.show()
```



In [164…

```
missing_percent = df.isnull().mean()

cols_to_drop = missing_percent[missing_percent > 0.7].index
df.drop(columns=cols_to_drop, inplace=True)
```

In [165…

```
for col in ['issue_d', 'earliest_cr_line', 'last_pymnt_d']:
    if col in df.columns:
        df[col] = pd.to_datetime(df[col], format='%b-%Y', errors='coerce')
```

In [166…

```
for col in df.select_dtypes(include=['float64']):
    df[col] = df[col].astype('float32')

for col in df.select_dtypes(include=['int64']):
    df[col] = df[col].astype('int32')
```

In [167…

```
df['loan_status'].value_counts(normalize=True)
```

Out[167…

```
loan_status
Current                                                0.678153
Fully Paid                                             0.234086
Charged Off                                            0.050991
Late (31-120 days)                                     0.013062
Issued                                                 0.009534
In Grace Period                                        0.007047
Late (16-30 days)                                      0.002656
Does not meet the credit policy. Status:Fully Paid     0.002240
Default                                                0.001374
Does not meet the credit policy. Status:Charged Off    0.000858
Name: proportion, dtype: float64
```

In [168…

```
df = df[df['loan_status'].isin(['Fully Paid', 'Charged Off'])]
```

In [169…

```
df['target'] = df['loan_status'].map({
    'Fully Paid': 0,
    'Charged Off': 1
})
```

In [170…

```
df['target'].value_counts(normalize=True)
```

Out[170…

```
target
0    0.821134
1    0.178866
Name: proportion, dtype: float64
```

In [171…

```
def compute_woe_iv(df, feature, target, bins=10):
    df_temp = df[[feature, target]].copy()

    # Create bins
    df_temp['bin'] = pd.qcut(df_temp[feature], q=bins, duplicates='drop')

    # Group by bin
    grouped = df_temp.groupby('bin')[target].agg(['count', 'sum'])
    grouped.columns = ['total', 'bad']
    grouped['good'] = grouped['total'] - grouped['bad']

    # Distributions
    grouped['bad_dist'] = grouped['bad'] / grouped['bad'].sum()
    grouped['good_dist'] = grouped['good'] / grouped['good'].sum()

    # WOE
    grouped['woe'] = np.log(grouped['good_dist'] / grouped['bad_dist'])

    # IV
    grouped['iv'] = (grouped['good_dist'] - grouped['bad_dist']) * grouped['woe'

    iv = grouped['iv'].sum()

    return grouped, iv
```

In [172…

```
woe_table, iv_value = compute_woe_iv(df, 'int_rate', 'target')

print("IV:", iv_value)
woe_table
```

```
IV: 0.40513808817729324
```

```
C:\Users\pulsa\AppData\Local\Temp\ipykernel_14312\1313088627.py:8: FutureWarning:
The default of observed=False is deprecated and will be changed to True in a futu
re version of pandas. Pass observed=False to retain current behavior or observed=
True to adopt the future default and silence this warning.
  grouped = df_temp.groupby('bin')[target].agg(['count', 'sum'])
```

Out[172…

bin	total	bad	good	bad_dist	good_dist	woe	iv
(5.319, 7.9]	31640	1700	29940	0.037571	0.144134	1.344520	0.143277
(7.9, 9.76]	19199	1676	17523	0.037040	0.084358	0.823057	0.038945
(9.76, 11.14]	25572	2793	22779	0.061726	0.109660	0.574676	0.027546
(11.14, 12.49]	27272	3575	23697	0.079009	0.114080	0.367336	0.012883
(12.49, 13.53]	23331	3549	19782	0.078434	0.095233	0.194060	0.003260
(13.53, 14.49]	26313	4653	21660	0.102833	0.104273	0.013908	0.000020
(14.49, 15.8]	25421	5286	20135	0.116823	0.096932	-0.186649	0.003713
(15.8, 17.57]	25963	6478	19485	0.143167	0.093803	-0.422814	0.020872
(17.57, 19.52]	23849	6640	17209	0.146747	0.082846	-0.571726	0.036534
(19.52, 28.99]	24411	8898	15513	0.196650	0.074681	-0.968195	0.118089

In [173…

```python
iv_dict = {}

num_cols = df.select_dtypes(include=['float32', 'float64', 'int32']).columns

for col in num_cols:
    if col != 'target':
        try:
            _, iv = compute_woe_iv(df, col, 'target')
            iv_dict[col] = iv
        except:
            pass

iv_df = pd.DataFrame(iv_dict.items(), columns=['feature', 'IV'])
iv_df.sort_values(by='IV', ascending=False).head(20)
```

```
C:\Users\pulsa\AppData\Local\Temp\ipykernel_14312\1313088627.py:8: FutureWarning:
The default of observed=False is deprecated and will be changed to True in a futu
re version of pandas. Pass observed=False to retain current behavior or observed=
True to adopt the future default and silence this warning.
  grouped = df_temp.groupby('bin')[target].agg(['count', 'sum'])
C:\Users\pulsa\AppData\Local\Temp\ipykernel_14312\1313088627.py:8: FutureWarning:
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C:\Users\pulsa\AppData\Local\Temp\ipykernel_14312\1313088627.py:8: FutureWarning:
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C:\Users\pulsa\AppData\Local\Temp\ipykernel_14312\1313088627.py:8: FutureWarning:
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C:\Users\pulsa\AppData\Local\Temp\ipykernel_14312\1313088627.py:8: FutureWarning:
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re version of pandas. Pass observed=False to retain current behavior or observed=
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C:\Users\pulsa\AppData\Local\Temp\ipykernel_14312\1313088627.py:8: FutureWarning:
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re version of pandas. Pass observed=False to retain current behavior or observed=
True to adopt the future default and silence this warning.
  grouped = df_temp.groupby('bin')[target].agg(['count', 'sum'])
```

```
C:\Users\pulsa\AppData\Local\Temp\ipykernel_14312\1313088627.py:8: FutureWarning: The default of observed=False is deprecated and will be changed to True in a future version of pandas. Pass observed=False to retain current behavior or observed=True to adopt the future default and silence this warning.
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  grouped = df_temp.groupby('bin')[target].agg(['count', 'sum'])
```

```
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C:\Users\pulsa\AppData\Local\Temp\ipykernel_14312\1313088627.py:8: FutureWarning:
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C:\Users\pulsa\AppData\Local\Temp\ipykernel_14312\1313088627.py:8: FutureWarning:
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re version of pandas. Pass observed=False to retain current behavior or observed=
True to adopt the future default and silence this warning.
  grouped = df_temp.groupby('bin')[target].agg(['count', 'sum'])
```

Out[173...

	feature	IV
26	last_pymnt_amnt	4.990293
21	total_rec_prncp	3.157258
19	total_pymnt	1.508360
20	total_pymnt_inv	1.494790
5	int_rate	0.405138
8	dti	0.082735
15	revol_util	0.068749
7	annual_inc	0.060524
0	id	0.059433
1	member_id	0.059040
31	tot_cur_bal	0.044989
4	funded_amnt_inv	0.025981
32	total_rev_hi_lim	0.025439
3	funded_amnt	0.025207
2	loan_amnt	0.024632
6	installment	0.020347
10	inq_last_6mths	0.016499
22	total_rec_int	0.015958
16	total_acc	0.007545
14	revol_bal	0.005549

In [174...
```
df.drop(columns=['last_pymnt_amnt'], inplace=True)
```

In [175...
```
df.drop(columns=['total_rec_late_fee'], inplace=True)
```

Findings:

int_rate → 0.405 🔥 (very strong) dti → 0.08 👍 revol_util → 0.068 👍 annual_inc → 0.060 👍

In [176...
```
df_temp = df[['int_rate', 'target']].copy()

df_temp['bin'] = pd.qcut(df_temp['int_rate'], q=10, duplicates='drop')

grouped = df_temp.groupby('bin', observed=False)['target'].agg(['count', 'sum'])
```

In [177...
```
iv_df.sort_values(by='IV', ascending=False).head(10)
```

Out[177…

	feature	IV
26	last_pymnt_amnt	4.990293
21	total_rec_prncp	3.157258
19	total_pymnt	1.508360
20	total_pymnt_inv	1.494790
5	int_rate	0.405138
8	dti	0.082735
15	revol_util	0.068749
7	annual_inc	0.060524
0	id	0.059433
1	member_id	0.059040

In [178…

```
def apply_woe(df, feature, target, bins=10):
    df_temp = df[[feature, target]].copy()

    df_temp['bin'] = pd.qcut(df_temp[feature], q=bins, duplicates='drop')

    grouped = df_temp.groupby('bin', observed=False)[target].agg(['count', 'sum'
    grouped.columns = ['total', 'bad']
    grouped['good'] = grouped['total'] - grouped['bad']

    # Avoid division by zero
    grouped['bad_dist'] = grouped['bad'] / grouped['bad'].sum()
    grouped['good_dist'] = grouped['good'] / grouped['good'].sum()

    grouped['woe'] = np.log((grouped['good_dist'] + 1e-5) / (grouped['bad_dist']

    woe_map = grouped['woe'].to_dict()

    df[feature + '_woe'] = pd.qcut(df[feature], q=bins, duplicates='drop').map(w

    return df
```

In [179…

```
top_features = ['int_rate', 'dti', 'revol_util', 'annual_inc', 'tot_cur_bal']

for col in top_features:
    df = apply_woe(df, col, 'target')
```

In [180…

```
woe_cols = [col + '_woe' for col in top_features]

X = df[woe_cols]
y = df['target']
```

In [181…

```
from sklearn.model_selection import train_test_split

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
```

In [182…
```
X.isnull().sum()
```

Out[182…
```
int_rate_woe            0
dti_woe                 0
revol_util_woe        199
annual_inc_woe          0
tot_cur_bal_woe     63708
dtype: int64
```

In [183…
```
top_features = ['int_rate', 'dti', 'revol_util', 'annual_inc', 'tot_cur_bal']

for feature in top_features:
    df = apply_woe(df, feature, 'target')
```

In [184…
```
woe_cols = [col + '_woe' for col in top_features]

X = df[woe_cols].astype(float)
y = df['target']
```

In [185…
```
X = X.fillna(0)
```

In [186…
```
X.isnull().sum()
```

Out[186…
```
int_rate_woe        0
dti_woe             0
revol_util_woe      0
annual_inc_woe      0
tot_cur_bal_woe     0
dtype: int64
```

In [187…
```
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

model = LogisticRegression(max_iter=1000)
model.fit(X_train, y_train)
```

Out[187…

LogisticRegression

LogisticRegression(max_iter=1000)

In [188…
```
from sklearn.metrics import roc_auc_score

y_prob = model.predict_proba(X_test)[:, 1]

print("ROC-AUC:", roc_auc_score(y_test, y_prob))
```

```
ROC-AUC: 0.6918034292422436
```

In [189…
```
from scipy.stats import ks_2samp

ks = ks_2samp(
    y_prob[y_test == 1],
    y_prob[y_test == 0]
)
```

```
print("KS:", ks.statistic)
```

KS: 0.2769623709716786

In [190…

```
import numpy as np

# Example scaling
factor = 20 / np.log(2)
offset = 600 - factor * np.log(50)

score = offset - factor * np.log(y_prob / (1 - y_prob))
```

In [191…

```
df_test = X_test.copy()
df_test['score'] = score
df_test['target'] = y_test.values
```

In [192…

```
import numpy as np

# Standard settings
PDO = 20   # Points to Double Odds
base_score = 600
base_odds = 50   # odds = good:bad

factor = PDO / np.log(2)
offset = base_score - factor * np.log(base_odds)
```

In [193…

```
y_prob = model.predict_proba(X_test)[:, 1]

score = offset - factor * np.log(y_prob / (1 - y_prob))
```

In [194…

```
scorecard_df = X_test.copy()
scorecard_df['target'] = y_test.values
scorecard_df['prob_default'] = y_prob
scorecard_df['score'] = score
```

In [195…

```
scorecard_df['score'].describe()
```

Out[195…

```
count    50595.000000
mean       535.971113
std         21.779025
min        484.753420
25%        519.766039
50%        534.315281
75%        550.451397
max        593.163933
Name: score, dtype: float64
```

In [196…

```
scorecard_df['risk_band'] = pd.qcut(
    scorecard_df['score'],
    q=5,
    labels=['Very High Risk', 'High Risk', 'Medium', 'Low Risk', 'Very Low Risk'
)
```

In [197…

```
scorecard_df.groupby('risk_band')['target'].mean()
```

```
C:\Users\pulsa\AppData\Local\Temp\ipykernel_14312\1405074136.py:1: FutureWarning:
The default of observed=False is deprecated and will be changed to True in a futu
re version of pandas. Pass observed=False to retain current behavior or observed=
True to adopt the future default and silence this warning.
  scorecard_df.groupby('risk_band')['target'].mean()
```

Out[197…

```
risk_band
Very High Risk    0.341140
High Risk         0.218599
Medium            0.163933
Low Risk          0.113252
Very Low Risk     0.059498
Name: target, dtype: float64
```

In [198…

```
def decision(score):
    if score > 700:
        return 'Approve'
    elif score > 650:
        return 'Review'
    else:
        return 'Reject'

scorecard_df['decision'] = scorecard_df['score'].apply(decision)
```

In [199…

```
loan_amount = 10000   # avg loan
interest_rate = 0.15  # 15% annual return
lgd = 0.6             # Loss Given Default (60%)
```

In [200…

```
scorecard_df['expected_profit'] = (
    (1 - scorecard_df['prob_default']) * interest_rate * loan_amount
    - scorecard_df['prob_default'] * lgd * loan_amount
)
```

In [201…

```
scorecard_df.groupby('risk_band')['expected_profit'].mean()
```

```
C:\Users\pulsa\AppData\Local\Temp\ipykernel_14312\1324467470.py:1: FutureWarning:
The default of observed=False is deprecated and will be changed to True in a futu
re version of pandas. Pass observed=False to retain current behavior or observed=
True to adopt the future default and silence this warning.
  scorecard_df.groupby('risk_band')['expected_profit'].mean()
```

Out[201…

```
risk_band
Very High Risk   -1013.455958
High Risk         -196.401000
Medium             274.203965
Low Risk           656.982398
Very Low Risk     1062.541301
Name: expected_profit, dtype: float64
```

In [202…

```
scorecard_df.groupby('risk_band')['target'].mean()
```

```
C:\Users\pulsa\AppData\Local\Temp\ipykernel_14312\1405074136.py:1: FutureWarning:
The default of observed=False is deprecated and will be changed to True in a futu
re version of pandas. Pass observed=False to retain current behavior or observed=
True to adopt the future default and silence this warning.
  scorecard_df.groupby('risk_band')['target'].mean()
```

Out[202…

```
risk_band
Very High Risk    0.341140
High Risk         0.218599
Medium            0.163933
Low Risk          0.113252
Very Low Risk     0.059498
Name: target, dtype: float64
```

In [203…

```
approved = scorecard_df[scorecard_df['score'] > 650]

total_profit = approved['expected_profit'].sum()
approval_rate = len(approved) / len(scorecard_df)

print("Total Profit:", total_profit)
print("Approval Rate:", approval_rate)
```

```
Total Profit: 0.0
Approval Rate: 0.0
```

In [204…

```
results = []

for cutoff in range(600, 750, 20):
    approved = scorecard_df[scorecard_df['score'] > cutoff]

    profit = approved['expected_profit'].sum()
    approval_rate = len(approved) / len(scorecard_df)

    results.append((cutoff, profit, approval_rate))

pd.DataFrame(results, columns=['cutoff', 'profit', 'approval_rate'])
```

Out[204…

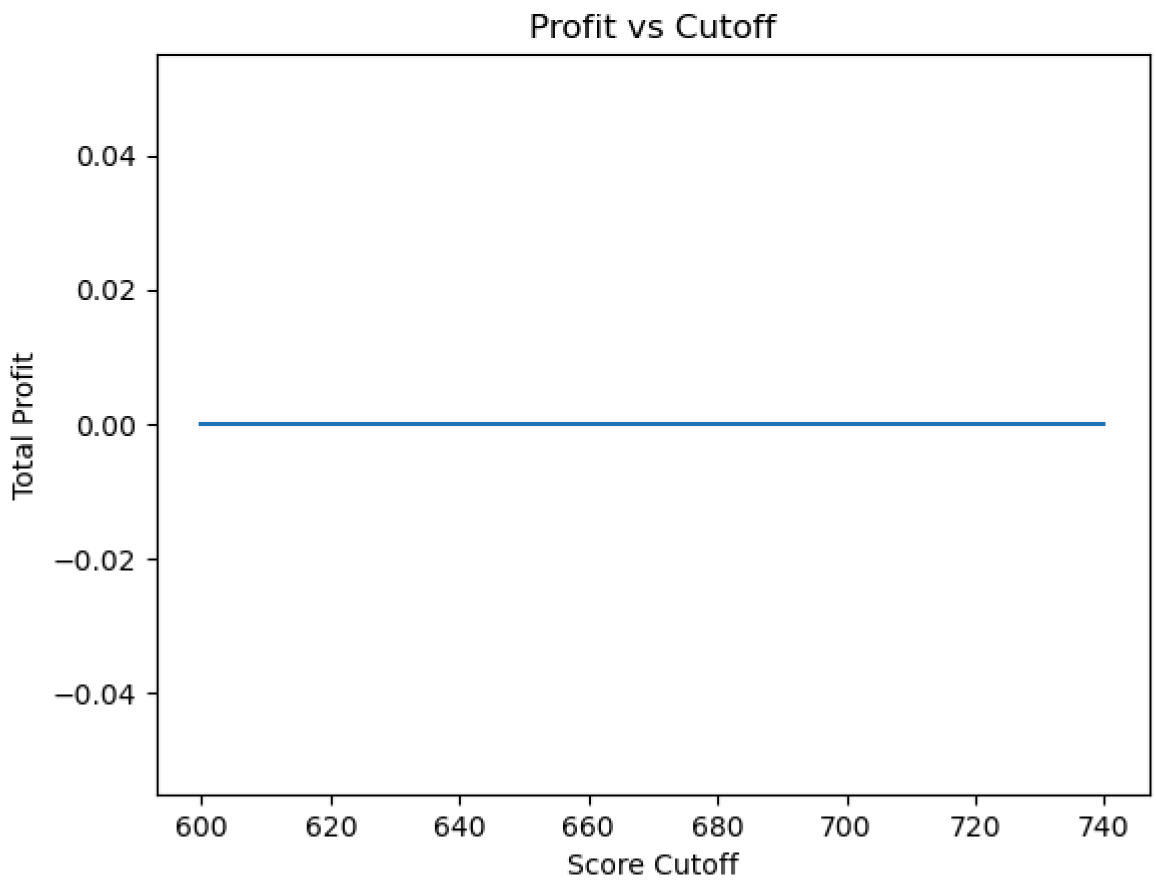
	cutoff	profit	approval_rate
0	600	0.0	0.0
1	620	0.0	0.0
2	640	0.0	0.0
3	660	0.0	0.0
4	680	0.0	0.0
5	700	0.0	0.0
6	720	0.0	0.0
7	740	0.0	0.0

In [205…

```
import matplotlib.pyplot as plt

res_df = pd.DataFrame(results, columns=['cutoff', 'profit', 'approval_rate'])

plt.plot(res_df['cutoff'], res_df['profit'])
plt.xlabel("Score Cutoff")
plt.ylabel("Total Profit")
plt.title("Profit vs Cutoff")
plt.show()
```



In [206…
```
scorecard_df['score'].describe()
```

Out[206…
```
count    50595.000000
mean       535.971113
std         21.779025
min        484.753420
25%        519.766039
50%        534.315281
75%        550.451397
max        593.163933
Name: score, dtype: float64
```

In [207…
```
PDO = 20
base_score = 650
base_odds = 20   # less aggressive

factor = PDO / np.log(2)
offset = base_score - factor * np.log(base_odds)
```

In [208…
```
score = offset - factor * np.log(y_prob / (1 - y_prob))
scorecard_df['score'] = score
```

In [209…
```
scorecard_df['score'].describe()
```

Out[209…

```
count    50595.000000
mean       612.409675
std         21.779025
min        561.191982
25%        596.204601
50%        610.753842
75%        626.889959
max        669.602495
Name: score, dtype: float64
```

In [210…

```
min_score = int(scorecard_df['score'].min())
max_score = int(scorecard_df['score'].max())

results = []

for cutoff in range(min_score, max_score, 20):
    approved = scorecard_df[scorecard_df['score'] > cutoff]

    profit = approved['expected_profit'].sum()
    approval_rate = len(approved) / len(scorecard_df)

    results.append((cutoff, profit, approval_rate))

pd.DataFrame(results, columns=['cutoff', 'profit', 'approval_rate'])
```

Out[210…

	cutoff	profit	approval_rate
0	561	7.931199e+06	1.000000
1	581	1.288336e+07	0.935132
2	601	2.009490e+07	0.670224
3	621	1.552018e+07	0.330823
4	641	7.075991e+06	0.122067
5	661	8.026322e+05	0.012452

Cutoff	Profit	Approval Rate
561	7.9M	100%
581	12.8M	93.5%
601	**20.0M 🔥 (MAX)**	**67.0%**
621	15.5M	33.0%
641	7.0M	12.2%
661	0.8M	1.2%

In [211…

```
y_pred = (scorecard_df['score'] > 600).astype(int)
```

In [212…

```
from sklearn.metrics import classification_report

print(classification_report(y_test, y_pred))
```

```
              precision    recall  f1-score   support

           0       0.70      0.27      0.39     41524
           1       0.12      0.47      0.20      9071

    accuracy                           0.30     50595
   macro avg       0.41      0.37      0.29     50595
weighted avg       0.59      0.30      0.35     50595
```

📌 Interpretation Guide Metric Meaning - [Precision How many predicted defaults are actually default] [Recall How many actual defaults you caught] [F1 Balance of precision & recall]

In [214…

```
from sklearn.metrics import recall_score, f1_score

recall = recall_score(y_test, y_pred)
f1 = f1_score(y_test, y_pred)

print("Recall:", recall)
print("F1 Score:", f1)
```

```
Recall: 0.4726050049608643
F1 Score: 0.19560605023612346
```

✅ Recall (Default = Class 1) Recall: 0.47 👉 You are catching ~47% of defaulters ⚠️ Precision (Default) Precision: 0.12 👉 Only 12% of predicted defaults are actually default ❌ Many false positives (rejecting good customers)F1: 0.20 👉 Low because: Precision is very low INSIGHT 👉 Your cutoff (~600) is too aggressive optimized for: ✔️ Profit ❌ Not classification balance ⚖️ TRADEOFF (VERY IMPORTANT) Goal What to optimize Profit Your current cutoff (~600) Risk control Higher recall Customer growth Higher precision

In [215…

```
for thresh in [580, 590, 600, 610]:
    y_pred = (scorecard_df['score'] > thresh).astype(int)

    print("Threshold:", thresh)
    print("Recall:", recall_score(y_test, y_pred))
    print("F1:", f1_score(y_test, y_pred))
    print()
```

```
Threshold: 580
Recall: 0.8668283540954691
F1: 0.2769637196195844

Threshold: 590
Recall: 0.6799691323999559
F1: 0.2387967246752744

Threshold: 600
Recall: 0.4726050049608643
F1: 0.19560605023612346

Threshold: 610
Recall: 0.2870686804100981
F1: 0.1485241693996863
```

In [216…

```
for p in [0.15, 0.2, 0.25, 0.3]:
    y_pred = (y_prob > p).astype(int)

    print("Threshold:", p)
```

```
    print("Recall:", recall_score(y_test, y_pred))
    print("F1:", f1_score(y_test, y_pred))
```

```
Threshold: 0.15
Recall: 0.7662881710946974
F1: 0.3772386844676001
Threshold: 0.2
Recall: 0.596736853709624
F1: 0.3869330569355588
Threshold: 0.25
Recall: 0.4332488149046412
F1: 0.37489268339215875
Threshold: 0.3
Recall: 0.27736743468195346
F1: 0.31821918674508315
```

In [219…

```
features = ['int_rate', 'dti', 'revol_util', 'annual_inc', 'tot_cur_bal']

X = df[features]
y = df['target']
```

In [220…

```
X = X.fillna(-999)
```

In [221…

```
pip install lightgbm
```

```
Requirement already satisfied: lightgbm in c:\users\pulsa\anaconda3\lib\site-pack
ages (4.6.0)
Requirement already satisfied: numpy>=1.17.0 in c:\users\pulsa\anaconda3\lib\site
-packages (from lightgbm) (1.26.4)
Requirement already satisfied: scipy in c:\users\pulsa\anaconda3\lib\site-package
s (from lightgbm) (1.16.3)
Note: you may need to restart the kernel to use updated packages.
```

In [222…

```
import lightgbm as lgb
from sklearn.model_selection import train_test_split
from sklearn.metrics import roc_auc_score

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

model_lgb = lgb.LGBMClassifier(
    n_estimators=200,
    learning_rate=0.05,
    max_depth=5,
    class_weight='balanced'
)

model_lgb.fit(X_train, y_train)

y_prob_lgb = model_lgb.predict_proba(X_test)[:, 1]

print("LightGBM ROC-AUC:", roc_auc_score(y_test, y_prob_lgb))
```

```
[LightGBM] [Info] Number of positive: 36177, number of negative: 166199
[LightGBM] [Info] Auto-choosing row-wise multi-threading, the overhead of testing
was 0.001602 seconds.
You can set `force_row_wise=true` to remove the overhead.
And if memory is not enough, you can set `force_col_wise=true`.
[LightGBM] [Info] Total Bins 1244
[LightGBM] [Info] Number of data points in the train set: 202376, number of used
features: 5
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.500000 -> initscore=-0.000000
[LightGBM] [Info] Start training from score -0.000000
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
LightGBM ROC-AUC: 0.6976057220451986
```

In [224…

```
pip install catboost
```

```
Requirement already satisfied: catboost in c:\users\pulsa\anaconda3\lib\site-pack
ages (1.2.10)
Requirement already satisfied: graphviz in c:\users\pulsa\anaconda3\lib\site-pack
ages (from catboost) (0.21)
Requirement already satisfied: matplotlib in c:\users\pulsa\anaconda3\lib\site-pa
ckages (from catboost) (3.8.0)
Requirement already satisfied: numpy<3.0,>=1.16.0 in c:\users\pulsa\anaconda3\lib
\site-packages (from catboost) (1.26.4)
Requirement already satisfied: pandas<4.0,>=0.24 in c:\users\pulsa\anaconda3\lib
\site-packages (from catboost) (2.3.3)
Requirement already satisfied: scipy in c:\users\pulsa\anaconda3\lib\site-package
s (from catboost) (1.16.3)
Requirement already satisfied: plotly in c:\users\pulsa\anaconda3\lib\site-packag
es (from catboost) (5.9.0)
Requirement already satisfied: six in c:\users\pulsa\appdata\roaming\python\pytho
n311\site-packages (from catboost) (1.16.0)
Requirement already satisfied: python-dateutil>=2.8.2 in c:\users\pulsa\appdata\r
oaming\python\python311\site-packages (from pandas<4.0,>=0.24->catboost) (2.9.0.p
ost0)
Requirement already satisfied: pytz>=2020.1 in c:\users\pulsa\anaconda3\lib\site-
packages (from pandas<4.0,>=0.24->catboost) (2023.3.post1)
Requirement already satisfied: tzdata>=2022.7 in c:\users\pulsa\anaconda3\lib\sit
e-packages (from pandas<4.0,>=0.24->catboost) (2023.3)
Requirement already satisfied: contourpy>=1.0.1 in c:\users\pulsa\anaconda3\lib\s
ite-packages (from matplotlib->catboost) (1.2.0)
Requirement already satisfied: cycler>=0.10 in c:\users\pulsa\anaconda3\lib\site-
packages (from matplotlib->catboost) (0.11.0)
Requirement already satisfied: fonttools>=4.22.0 in c:\users\pulsa\anaconda3\lib
\site-packages (from matplotlib->catboost) (4.25.0)
Requirement already satisfied: kiwisolver>=1.0.1 in c:\users\pulsa\anaconda3\lib
\site-packages (from matplotlib->catboost) (1.4.4)
Requirement already satisfied: packaging>=20.0 in c:\users\pulsa\appdata\roaming
\python\python311\site-packages (from matplotlib->catboost) (24.0)
Requirement already satisfied: pillow>=6.2.0 in c:\users\pulsa\anaconda3\lib\site
-packages (from matplotlib->catboost) (10.2.0)
Requirement already satisfied: pyparsing>=2.3.1 in c:\users\pulsa\anaconda3\lib\s
ite-packages (from matplotlib->catboost) (3.0.9)
Requirement already satisfied: tenacity>=6.2.0 in c:\users\pulsa\anaconda3\lib\si
te-packages (from plotly->catboost) (8.2.2)
Note: you may need to restart the kernel to use updated packages.
```

In [225…

```python
from catboost import CatBoostClassifier

model_cat = CatBoostClassifier(
    iterations=200,
    learning_rate=0.05,
    depth=5,
    verbose=0
)

model_cat.fit(X_train, y_train)

y_prob_cat = model_cat.predict_proba(X_test)[:, 1]

print("CatBoost ROC-AUC:", roc_auc_score(y_test, y_prob_cat))
```

```
CatBoost ROC-AUC: 0.6962363445080648
```

In [226…

```python
features = [
    'loan_amnt',
```

```
    'int_rate',
    'term',
    'grade',
    'sub_grade',
    'emp_length',
    'home_ownership',
    'annual_inc',
    'verification_status',
    'purpose',
    'dti',
    'delinq_2yrs',
    'revol_util',
    'total_acc',
    'tot_cur_bal'
]

X = df[features].copy()
y = df['target']
```

In [227…

```
X['term'] = X['term'].str.extract('(\d+)').astype(float)
```

In [228…

```
X['emp_length'] = X['emp_length'].str.extract('(\d+)').astype(float)
```

In [229…

```
X = X.fillna(-999)
```

In [230…

```
X['income_to_loan'] = X['annual_inc'] / (X['loan_amnt'] + 1)
X['dti_income'] = X['dti'] * X['annual_inc']
X['credit_util_ratio'] = X['revol_util'] / (X['total_acc'] + 1)
X['loan_to_income'] = X['loan_amnt'] / (X['annual_inc'] + 1)
```

In [231…

```
cat_features = [
    'grade',
    'sub_grade',
    'home_ownership',
    'verification_status',
    'purpose'
]
```

In [232…

```
from catboost import CatBoostClassifier
from sklearn.model_selection import train_test_split
from sklearn.metrics import roc_auc_score

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

model_cat = CatBoostClassifier(
    iterations=500,
    learning_rate=0.05,
    depth=6,
    eval_metric='AUC',
    verbose=100
)

model_cat.fit(
    X_train, y_train,
    cat_features=cat_features,
    eval_set=(X_test, y_test)
```

```
)

y_prob_cat = model_cat.predict_proba(X_test)[:, 1]

print("CatBoost ROC-AUC:", roc_auc_score(y_test, y_prob_cat))
```

```
0:      test: 0.6855202 best: 0.6855202 (0)     total: 226ms    remaining: 1m 52s
100:    test: 0.7067348 best: 0.7067348 (100)   total: 24.3s    remaining: 1m 35s
200:    test: 0.7087117 best: 0.7087117 (200)   total: 47.6s    remaining: 1m 10s
300:    test: 0.7101615 best: 0.7101615 (300)   total: 1m 10s   remaining: 46.5s
400:    test: 0.7113278 best: 0.7113310 (399)   total: 1m 31s   remaining: 22.6s
499:    test: 0.7120652 best: 0.7120652 (499)   total: 1m 53s   remaining: 0us

bestTest = 0.7120652086
bestIteration = 499

CatBoost ROC-AUC: 0.7120652086174879
```